

Abdul Basit Sajid

Backend / Infrastructure Engineer

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SUMMARY

Backend and infrastructure engineer, 3 years, remote from Pakistan for a UK company since 2023. Strongest work is capacity and latency engineering on a real-time GPU inference backend, and the cost analysis that argued against building the system that would have scaled it. Production experience is Python and Node on GCP/AWS; the Go, Kubernetes and Postgres-internals work below is open source.

EXPERIENCE

Software Engineer, EliteITTeam, Manchester UK (Remote)

Sept 2024 – Present

- Took a real-time multi-model computer-vision backend from a measured ~ 3 concurrent users to a proved capacity of 700 concurrent under load test: 99.4% frame delivery, 132 ms p50, \$0.0106 per user-hour.
- Cut per-frame inference 2,800 ms $\rightarrow$ 60 ms ($46\times$): four YOLO models onto CUDA, an $O(n^2)$ Python NMS loop replaced with `torchvision.ops.nms`, and per-detection GPU $\rightarrow$ CPU transfers batched into one copy.
- Cut p50 871 ms $\rightarrow$ 59 ms by removing a `broadcast=True` fan-out that made every response $O(N)$ in connected sockets, and pinning each session to one container via an nginx hash. Found in a 7-configuration sweep in which three of my own hypotheses were falsified, and the only change that moved the ceiling rather than the tail.
- Wrote the capacity decision document for a 60,000-user target: corrected an internal \$206k/mo projection to \$475–570k/mo from measured per-box capacity, established that GPUs are excluded from GCP's Flexible CUD, and recommended **not** building the interim GPU fleet. The adopted on-device path removes GPU inference cost entirely.

Associate Software Engineer, EliteITTeam, Manchester UK (Remote)

Sept 2023 – Sept 2024

- Root-caused a Stripe webhook failure and repaired the affected production data with a dry-run pass, row checksums and a written rollback procedure before executing against live records.
- Owned production migrations on a learning platform end to end (provider cutovers and backup/restore procedures rehearsed against a staging copy before each live run), and took a reporting endpoint 2.5 s $\rightarrow$ 650 ms by collapsing six sequential MongoDB queries into one `$facet` aggregation.

OPEN-SOURCE INFRASTRUCTURE PROJECTS

pgbranch: copy-on-write Postgres branching (Go, OverlayFS, Docker)

github.com/abd-ulbasit/pgbranch

- Diagnosed why branching a 5 GiB database took 61.9 s and left a 5.05 GiB writable layer, a full copy inside a system whose premise is that it copies nothing. Postgres's pre-WAL-replay `SyncDataDirectory` opens every PGDATA file `O_RDWR`, and on OverlayFS an `O_RDWR` open of a lower-layer file forces whole-file copy-up.
- Proved the mechanism instead of arguing it: a single-variable control run with `recovery_init_sync_method=syncfs` ended at 16 KiB of writable layer, not 5.05 GiB. The one-flag fix takes create to 1.89 s / 33 MiB.
- Published benchmarks keep the pre-fix numbers that contradicted the project's thesis, and say why they are not comparable.

upgradescope: Kubernetes upgrade-readiness scanner (Go, client-go)

github.com/abd-ulbasit/upgradescope

- Collectors degrade instead of failing: a 401/403 from a managed control plane's metrics endpoint becomes a named unavailable capability, so the report states which signal was lost rather than reporting a clean scan.
- Argued against a controller-runtime reconciler: the verdict is a function of (inventory, knowledge base, target) and two of those three change with no Kubernetes event, so a watch-driven loop requeues on unrelated pod churn yet still misses the day a component's EOL passes.
- The API-lifecycle dataset is generated and CI-gated, plus a check that diffs `go list k8s.io/api/...` against the generator's imports so a new API group cannot silently pass.

goqueue: distributed message queue (Go)

github.com/abd-ulbasit/goqueue

- Found a bucket-drain race in a hierarchical timing wheel: an element pointer held across the callback unlock is silently invalidated by a concurrent cancel, truncating the traversal and stranding every remaining timer in that bucket for up to 7.76 days.
- Invisible to `-race` (every access really was under the mutex) and to the existing tests; wrote deterministic regression tests that fail against the old implementation, then fixed it.

forgepoint: ML-lifecycle platform (Go, Kubernetes)

github.com/abd-ulbasit/forgepoint

Takes a model from training through serving and back to retraining when it drifts, as a closed loop. Eleven services, each owning its own database, have to agree on what happened without a distributed transaction. A cancelled workflow still has to durably record the cancellation on a context that is already dead.

WRITING, EDUCATION & CERTIFICATIONS

Bookstore Kubernetes Guide

abd-ulbasit.github.io/bookstore-kubernetes-guide

16 parts, 115 chapters: one application from a container to a production cluster.

BS Computer Science, NUST Islamabad | AWS Certified Solutions Architect (Associate), July 2025

SKILLS

Languages: Go, Python, TypeScript, SQL, Rust | **Data:** PostgreSQL, MongoDB, Redis | **Cloud:** AWS, GCP

Infrastructure: Kubernetes, Docker, Helm, Terraform, Nginx, GitHub Actions, Prometheus, gRPC, NATS JetStream

Practices: Distributed systems, concurrency, load testing and capacity planning, cost modelling, production migrations